

Table of Contents

- [UW12GNoiseSrcDefTopic](#)
- [UW12GNoiseSrc_About.1.2](#)
- [UW12GNoiseSrc_About.1.3](#)
- [UW12GNoiseSrc_About.1.4](#)
- [UW12GNoiseSrc_Displays.4.1](#)
- [UW12GNoiseSrc_Displays.4.2](#)
- [UW12GNoiseSrc_Prog.3.1](#)
- [UW12GNoiseSrc_Prog.3.2](#)
- [UW12GNoiseSrc_Prog.3.3](#)
- [UW12GNoiseSrc_Prog.3.4](#)
- [UW12GNoiseSrc_TOO](#)

UW12GNoiseSrcDefTopic

<

>

About the UltraWave 12G MWNoiseSource

About the UltraWave 12G MWNoiseSource

The MWNoiseSource is a wideband noise source that generates a calibrated excess noise ratio (ENR). The primary application of the MWNoiseSource is noise figure tests.

This section includes the following topics:

- [Features](#)

•	Safety Information
---	------------------------------------

•	Key Terms
---	---------------------------

MWNoiseSource information includes the following sections:

•	UltraWave 12G MWNoiseSource Theory of Operation
---	---

•	UltraWave 12G MWNoiseSource Programming
---	---

•	UltraWave 12G MWNoiseSource Debug Display
---	---

Related Topics

•	UltraWave 12G Specifications
---	--

•	MWNoiseSource (VBT syntax)
---	--

•	UltraWave 12G DIB Design Guidelines
---	---

•	About the UltraWave 12G Subsystem
---	---

•	UltraWave 12G Examples
---	--

	
---	---

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

UW12GNoiseSrc_About.1.2

<	>
-----------------	-----------------

About the UltraWave 12G MWNoiseSource : Features

Features

The MWNoiseSource generates wideband noise at a calibrated ENR.

Basic functional specifications are:

- | | |
|---|---------------------------------|
| • | Frequency range: 50MHz to 12GHz |
|---|---------------------------------|
- | | |
|---|---------------------------------|
| • | Excess Noise Ratio (ENR) level: |
| – | 50MHZ to 2.5GHz: >23dB |
| – | 2.5GHz to 6GHz: >23dB |
| – | 6GHz to 12GHz: >18dB |
- | | |
|---|---|
| • | Adjustable ENR for different device test requirements |
|---|---|

ENR accuracy is provided by an external calibration performed in the maintenance environment.
For UltraWave 12G hardware specifications, see UltraWave 12G Specifications.

<	>
-----------------	-----------------

--	--	--

UW12GNoiseSrc_About.1.3

<	>
-----------------	-----------------

About the UltraWave 12G MWNoiseSource : Safety Information

Safety Information

When operating Microwave instruments, always follow general UltraFLEX test system safety guidelines. See [Test System Safety](#).

<	>
-----------------	-----------------

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

UW12GNoiseSrc_About.1.4

<	>
-----------------	-----------------

About the UltraWave 12G MWNoiseSource : Key Terms

Key Terms

cold	Cold temperature condition simulated by a noise source switched off.
ENR	Excess noise ratio. The power of a noise source when it is on relative to when it is off.
hot	Hot temperature condition simulated by a noise source switched on.
noise figure	Noise added to the signal by a DUT.
target ENR	Desired target ENR for a calibrated ENR measurement. IG-XL selects a step attenuator setting to best approximate the target ENR and returns the actual calibrated ENR.
target ENR frequency	Frequency of interest for target ENR.

For more MW terms and definitions, see [Key Terms](#) in About the UltraWave 12G Subsystem.

<	>
-----------------	-----------------

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

UW12GNoiseSrc_Displays.4.1

<

>

UltraWave 12G MWNoiseSource Debug Display

UltraWave 12G MWNoiseSource Debug Display

The UltraWave 12G Microwave Debug Display can display one or more instrument blocks for the MWNoiseSource logical instrument, each showing the state of a MWNoiseSource in the context of a site.

This section includes the following topic:

- [UltraWave 12G MWNoiseSource Debug Display Controls](#)

For an overview of all displays supporting the UltraWave 12G subsystem, see [UltraWave 12G Displays](#).

For more information on using debug displays, see the following:

- [Using Debug Displays in TDE](#)

- [Producing Code from Debug Displays](#)

<

>

UW12GNoiseSrc_Displays.4.2

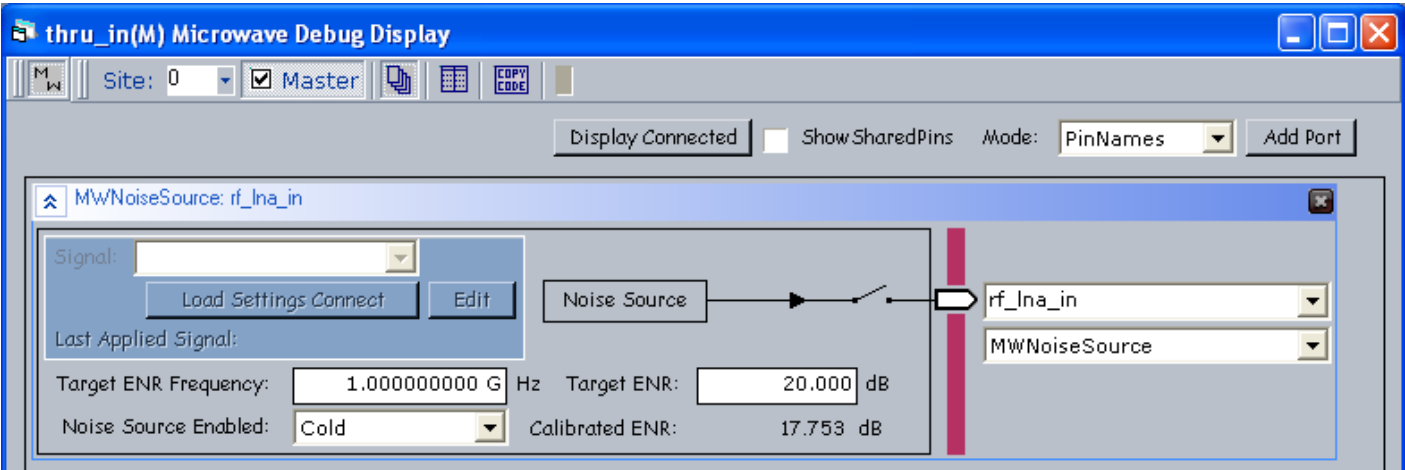


UltraWave 12G MWNoiseSource Debug Display : UltraWave 12G MWNoiseSource Debug Display Controls

UltraWave 12G MWNoiseSource Debug Display Controls

The display controls for the MWNoiseSource assigned to a port allow you to view and set some common MWNoiseSource parameters, which are also accessible through the Pins interface of the VBT language.

For information about the common debug display controls, see UltraWave 12G Displays.



MWNoiseSource Logical Instrument Display

The following controls are available on the MWNoiseSource logical instrument block:

Target ENR Frequency	Displays and sets the target frequency for the noise measurement. The tooltip displays the allowable frequency range.
Target ENR	Displays and sets the target ENR for the noise measurement. The tooltip displays the allowable target ENR range.
Noise Source Enabled	Displays and sets the noise source enabled mode: Cold or Hot.

Calibrated ENR	Displays the calibrated ENR value for the target ENR and frequency.
Signal	Displays the selected source Signal on a drop-down list of available signals.
Load Settings Connect	Loads the selected signal and connects the MWNoiseSource.
Info	Displays information for the selected Signal in a pop-up balloon window.
Last Applied Signal	Name of the Signal last loaded by the MWNoiseSource.
Connection relay	The relay control connects or disconnects the MWNoiseSource instrument to the specified pin or channel. Click the relay to open or close the connection.

<

>

2015 Teradyne, Inc. All rights reserved.

UW12GNoiseSrc_Prog.3.1

<

>

UltraWave 12G MWNoiseSource Programming

UltraWave 12G MWNoiseSource Programming

The MWNoiseSource is a simple instrument with two key operations used to make a noise figure measurement of a DUT by the Y-factor method. The MWNoiseSource provides:

- A Hot noise power source
- A Cold noise power source

The following topic is available:

- MWNoiseSource Pattern Control

For general aspects of programming instruments in the UltraWave 12G Subsystem, see [UltraWave 12G Instrument Programming](#).

For more information, see [MWNoiseSource](#) in the VBT Syntax Reference.

MWNoiseSource Reset Values

The table lists the local reset values for the MWNoiseSource.

MWNoiseSource Reset Values	
Setting	Default Value
Enabled Temperature	Cold
TargetENR	20dB
TargetENRFrequency	1GHz

Connections

Disconnected

<		>	
	2015 Teradyne, Inc. All rights reserved.		

UW12GNoiseSrc_Prog.3.2

<	>
-----------------	-----------------

UltraWave 12G MWNoiseSource Programming : MWNoiseSource Pattern Control

MWNoiseSource Pattern Control

You can control an MWNoiseSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming. You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution. The following topics are available:

- MWNoiseSource Pattern Microcodes
- MWNoiseSource ASCII Pattern Syntax

Note: To use pattern microcodes to control an MWNoiseSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

<	>
-----------------	-----------------

UW12GNoiseSrc_Prog.3.3

<

>

UltraWave 12G MWNoiseSource Programming : MWNoiseSource Pattern Control : MWNoiseSource Pattern Microcodes

MWNoiseSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWNoiseSource instrument from a pattern.

MWNoiseSource Pattern Microcodes	
Microcode	Functional Description
Disconnect	Instructs an MWNoiseSource to reset all connections to their disconnected state.
LoadSettingsConnect <SignalName>	Instructs an MWNoiseSource to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.

Note the following when using microcodes with an MWNoiseSource:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:

– LoadSettingsConnect: 200 - 400s (typical)

– Disconnect: <200s (typical)

- An MWNoiseSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.

- When using an UltraWave logical instrument (such as an MWNoiseSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.

– IG-XL returns an error if you do not follow this guideline.

– Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see [Using UltraWave 12G Instruments with POP](#).

UW12GNoiseSrc_Prog.3.4

	
---	---

UltraWave 12G MWNoiseSource Programming : [MWNoiseSource Pattern Control](#) : MWNoiseSource ASCII Pattern Syntax

MWNoiseSource ASCII Pattern Syntax

MWNoiseSource pattern microcodes are instructions that you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWNoiseSource to perform a particular action, such as starting a new signal.

You can only use MWNoiseSource microcodes with pins that the instruments statement has specified as MWNoiseSource. For example:

```
instruments = {
    DUT_AIN:MWNoiseSource;
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

If a vector contains any MWNoiseSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWNoiseSource microcode on a particular vector is:

(pin-list : MWNoiseSource = microcode)

The following code shows an example of ASCII pattern microcode syntax for an MWNoiseSource.

The figure shows this pattern in PatternTool.

```
instruments = {
    Noise:MWNoiseSource 1;
}
```

vm_vector

vm_MWNoiseSourceOnly_A (\$tset , Dig4)

```
{
    > BasicSource2 0 ;
```

16/22


```
halt
}
```

MWNoiseSourceOnly.PAT (vm_MWNoiseSourceOnly_A - VM)						
Tset	Command	Label	Vector	Noise (MWNoiseSource)	Di- g- 4	Comment
BasicSource2			0		0	
BasicSource2		start_label MWNoiseSourceOnly:	1		0	
BasicSource2			2	LoadSettingsConnect Hot	0	
BasicSource2	repeat 10000		3		0	1mS wait time
BasicSource2			4	LoadSettingsConnect Cold	0	
BasicSource2	repeat 10000		5		0	1mS wait time
BasicSource2			6	Disconnect	0	
BasicSource2	repeat 2000		7		0	200uS wait time
BasicSource2			8		0	
BasicSource2			9		0	
BasicSource2			10		0	

Vertical: 65 Vectors Vectors 0 to 10

MWNoiseSource Pattern Microcodes Example

<

>

2015 Teradyne, Inc. All rights reserved.

UW12GNoiseSrc_TOO

<	>
----------------------	----------------------

UltraWave 12G MWNoiseSource Theory of Operation

UltraWave 12G MWNoiseSource Theory of Operation

The MWNoiseSource instrument produces a noise signal with a calibrated ENR value. There is a noise source on each of the four front ends of a Measure board.

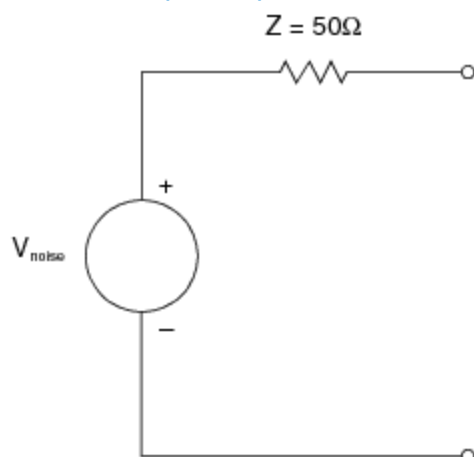
The following topics are available:

- [Noise Source Model](#)
- [ENR Adjustment](#)
- [MWNoiseSource Hardware Functions and Signal Flows](#)

For information on MWNoiseSource instrument performance specifications and standards, see [UltraWave 12G Specifications](#).

Noise Source Model

The MWNoiseSource can be modeled as an ideal broadband (white) noise source in series with a 50ohm output impedance. See the figure.



Equivalent Circuit Model of the MWNoiseSource

The noise source supports the Y-factor method for measuring the noise added by a device. The Y-factor method requires a noise source with a calibrated ENR. An accurate ENR value for the MWNoiseSource is calculated and stored during calibration.

The measurement technique based on the Y-factor method requires two measurements:

- Noise power for the cold condition

- Noise power for the hot condition

When the noise source is off, it simulates a cold temperature condition. When the noise source is switched on, it simulates a hot temperature condition.

ENR Adjustment

The UltraWave 12G noise source can provide different ENR levels to match device test requirements. The calibrated ENR is based on two programmable MWNoiseSource parameters:

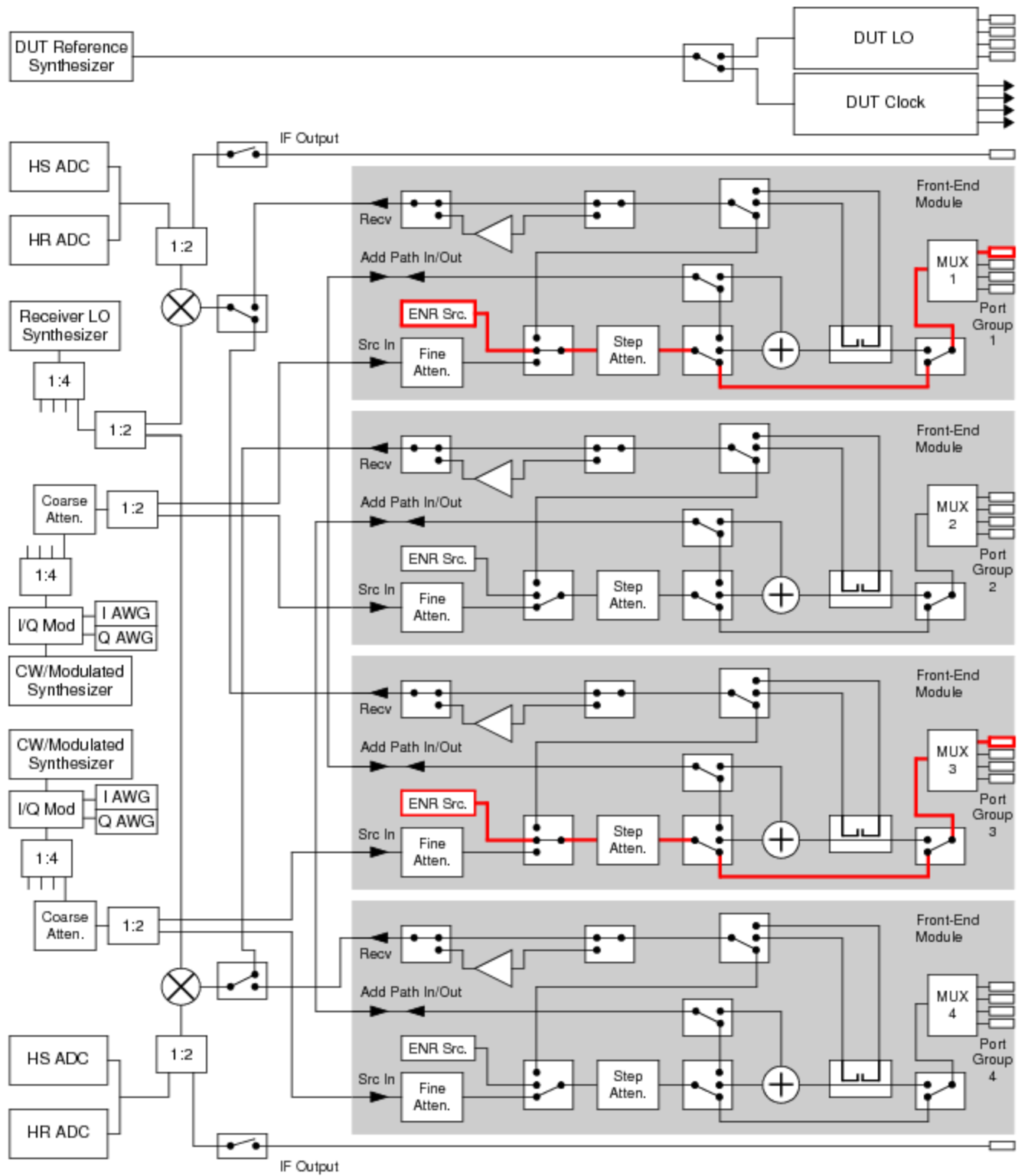
- Target ENR value: the desired ENR for the test

- Target ENR frequency: the frequency of interest for the test

Based on calibration data, the MWNoiseSource programs the step attenuator in the front end to set the calibrated ENR close to the target value. The actual calibrated ENR value for the specific port is returned for use in measurement calculations.

MWNoiseSource Hardware Functions and Signal Flows

The figure shows noise sources and signal paths to two ports simultaneously. The noise sources are not shared. Heavy lines represent signal flows.



UltraWave 12G MWNoiseSource Hardware Functions and Signal Flows

<

>

